Name = Babar Zareen
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subject = oop
Lab Task

QUESTION NO: 01

polymorphism overloading Calculator

code:

```java
import java.util.Scanner;  //name babar zareen sap id 74760

class Calculator {
    int sum(int a, int b) {
        return a + b;
    }

    int sum(int a, int b, int c) {
        return a + b + c;
    }

    float sum(float a, float b) {
        return a + b;
    }
}

class Exe {  //name babar zareen sap id 74760
    public static void main(String[] args) {
        int a, b, c;
        float x, y;

        Scanner s = new Scanner(System.in);
        Calculator calc = new Calculator();

        System.out.println("Enter values:");
```

```java
        System.out.print("Enter first integer: ");
        a = s.nextInt();
        System.out.print("Enter second integer: "); // name babar zareen sap id 74760
        b = s.nextInt();
        System.out.print("Enter third integer: ");
        c = s.nextInt();

        System.out.print("Enter first float: ");
        x = s.nextFloat();
        System.out.print("Enter second float: ");
        y = s.nextFloat();

        System.out.println("Sum of "+ a +" "+ b +" "+ c + " is " + calc.sum(a, b, c));
        System.out.println("Sum of " + a + " and " + b + " is " + calc.sum(a, b));
        System.out.println("Sum of " + x + " and " + y + " is " + calc.sum(x, y));

    }
}
```

Output:

```
Enter values:
Enter first integer: 12
Enter second integer: 10
Enter third integer: 13
Enter first float: 12.98
Enter second float: 32.987
Sum of 12 10 13 is 35
Sum of 12 and 10 is 22
Sum of 12.98 and 32.987 is 45.967


...Program finished with exit code 0
Press ENTER to exit console.
```

QUESTION NO: 02

polymorphism overriding BankAccount

code:

```java
// babar zareen sap id 74760
class BankAccount {
    String accountTitle;
    int accountNumber;

    void deposit(double amount) {
        System.out.println("Deposit in Bank Account: " + amount);
    }

    void withdraw(double amount) {
        System.out.println("Withdraw from Bank Account: " + amount);
    }
}

// babar zareen sap id 74760
class SavingAccount extends BankAccount {

    void deposit(double amount) {
        System.out.println("Deposit in Saving Account: " + amount);
    }

    void withdraw(double amount) {
        System.out.println("Withdraw from Saving Account: " + amount);
    }
}
```

```java
// babar zareen sap id 74760
class CheckingAccount extends BankAccount {
    void deposit(double amount) {
        System.out.println("Deposit in Checking Account: " + amount);
    }
    void withdraw(double amount) {
        System.out.println("Withdraw from Checking Account: " + amount);
    }
}
// babar zareen sap id 74760
public class Main {
    public static void main(String[] args) {

        BankAccount b;

        b = new SavingAccount();
        b.deposit(1000);
        b.withdraw(500);

        b = new CheckingAccount();
        b.deposit(2000);
        b.withdraw(800);
    }
}
```

output:

```
Deposit in Saving Account: 1000.0
Withdraw from Saving Account: 500.0
Deposit in Checking Account: 2000.0
Withdraw from Checking Account: 800.0


...Program finished with exit code 0
Press ENTER to exit console.
```

QUESTION NO: 03

quiz:

code:

```java
class Shape { //babar zareen sap id 74760
    void draw() {
        System.out.println("Drawing Shape");
    }
    double calculateArea() {
        return 0;
    }
}

class Circle extends Shape {
    double r;
    Circle(double r) { this.r = r; }

    void draw() {
        System.out.println("Drawing Circle");
    }

    double calculateArea() { // babar zareen sap id 74760
        return Math.PI * r * r;
    }
}
```

```java
class Cylinder extends Shape {
    double r, h;
    Cylinder(double r, double h) {
        this.r = r;
        this.h = h;
    }

    void draw() {
        System.out.println("Drawing Cylinder");
    }

    double calculateArea() {
        return 2 * Math.PI * r * (r + h);
    }
}
public class Main {
    public static void main(String[] args) { // babar zareen sap id 74760
        Circle c = new Circle(5);
        Cylinder cy = new Cylinder(3, 7);
        c.draw();
        System.out.println(c.calculateArea());
        cy.draw();
        System.out.println(cy.calculateArea());
    }
}
```

output:

```
Drawing Circle
78.53981633974483
Drawing Cylinder
188.49555921538757


...Program finished with exit code 0
Press ENTER to exit console.
```